Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Component

Meaning in my words How would you explain component to a friend?	**How can you tell?** If east and north are positive, which component is negative for a southeast trip?
Example or sketch What two arrows could you use to build a displacement of 3 m east and 4 m north?	**Non-example and why** A classmate counts the diagonal and both parts as three separate trips. What is counted twice?

Resultant

Meaning in my words How would you explain resultant to a friend?	**How can you tell?** Where does the single arrow start and end on a tip-to-tail drawing?
Example or sketch Walk 3 m east, then 4 m north. What single arrow joins your start and finish?	**Non-example and why** Someone labels the resultant 7 m. Which length did they add up?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Magnitude

Meaning in my words How would you explain magnitude to a friend?	**How can you tell?** For a displacement of 5 m west, what part gives the magnitude?
Example or sketch A vector has perpendicular parts of 3 m and 4 m. How long is the diagonal?	**Non-example and why** A partner gives the magnitude as −5 m. What is the minus sign doing?

Quadrant

Meaning in my words How would you explain quadrant to a friend?	**How can you tell?** Which way does an arrow point if its horizontal part is negative and its vertical part positive?
Example or sketch Sketch a vector with components +2 m and −3 m. Where is its tip relative to its tail?	**Non-example and why** Do two vectors with the same length always point into the same quadrant?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a right-angle journey; label x, y, and the resultant with its direction.



My marks mean

Send the whole movement

06

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Combine two displacement components into one arrow.

CHOOSE ONE JOB TOGETHER

NAVIGATION ENGINEER ☐

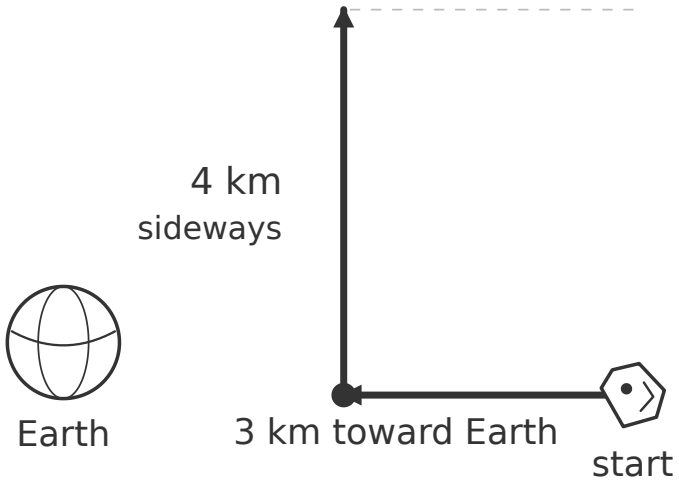
Draw one arrow for the whole displacement. Give its length and direction, and label both component legs.

MAP PRESENTER ☐

Use the map to report the whole movement. Explain why '**7 km** closer to Earth' is the wrong caption.

Point to a clue. Keep your first marks; add the update using the color key.

A tracker sees a **3 km** displacement toward Earth and **4 km** sideways. The second telescope needs the whole movement.



01 OUR FIRST CALL

What full movement should the second telescope track? Use your map.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

What part can the updated tracker measure? What part will it miss?

Switch speakers. Give the direction as an angle. Name the line you measure that angle from.

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks